

SOC ANALYST PROJECT REPORT

Report Title: PCAP Traffic Analysis – Malicious Network Activity Investigation

Analyst: Utibe Anthony

Date: 29th June, 2026

File Analyzed: 2017-12-23-traffic-analysis.pcap

Project Type: Personal SOC Analyst Portfolio Project

OBJECTIVE:

As part of my personal cybersecurity portfolio development, I conducted a full network traffic investigation on a PCAP file to identify Indicators of Compromise (IOCs), map attacker behavior to the MITRE ATT&CK framework, and document findings in a professional SOC analyst format.

TOOLS USED:

Tool	Purpose
Wireshark	PCAP analysis and packet inspection
VirusTotal	IP, domain and file hash reputation checking
AbuseIPDB	IP abuse and malicious activity verification
Shodan	IP infrastructure reconnaissance
Terminal/Bash/Sha256	File hashing and payload extraction

METHODOLOGY:

Created dedicated working directory /home/kali/pcap_analysis/

Copied original PCAP to preserve evidence integrity

Opened duplicate file in Wireshark for analysis

Investigated TCP streams, HTTP objects and conversation tables

Extracted and investigated all external IP addresses

Verified all suspicious IPs against VirusTotal, AbuseIPDB and Shodan

Mapped all discovered behaviors to MITRE ATT&CK framework.

FINDINGS:

Malicious & Suspicious External Ips

IP Address	Vendors Flagged	Role Identified
164.132.223.193	1 Vendor	Primary C2 server
103.192.0.2	2 Vendors	Ingress tool delivery
220.227.247.35	2 Vendors	HTTP data transfer-no filename
153.149.153.41	4 Vendors	Malware delivery-Emote linked
62.210.253.72	Unverified	Received 372 byte POST data
185.22.184.127	Unrated	Suspicious C2 over port 443
176.31.27.119	Unrated	Served suspicious CRL file

Internal Machines Compromised

Internal IP	Suspicious Activity Observed
172.16.67	Sent POST request to external C2 on port 8080
172.16.1.201	Communicated with a confirmed malicious IPs

Malicious Domain Identified

Domain	Vendors Flagged	Resolved IP
www.creomerx.co.jp	0 (domain clean)	153.149.153.41 (4 vendors flagged)

Suspicious Files & Data Transferred

File/Data	Size	Source IP	Observation
Msittwww2.crl	1,197 bytes	176.31.27.119	Fake certificate revocation list
Raw binary payload	372 bytes	153.149.153.41:8080	Encoded data-undetected by VirusTotal
POST data	372 bytes	62.210.253.72:8080	No filename-possible credential exfiltration
POST data	372 + 372 bytes	220.227.247.35	No filename-repeated pattern
Ingress payload	1,288 bytes	103.192.0.2	Delivered via ACK packet

Network Behaviors – IOCs

Conversation Settings

Name resolution

Absolute start time

Display raw data

Limit to display filter

Copy

Follow Stream...

Graph...

I/O Graphs

Protocol

Bluetooth

IPv7

Filter list for specific type

Ethernet - 20	IPv4 - 50	IPv6	TCP - 406	UDP - 140						
Address A	Port A	Address B	Port B	Packets	Bytes	Stream ID	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A
172.16.1.89	49199	185.22.184.127	443	35	11 kB	138	17	4 kB	18	7 kB
172.16.1.89	49200	185.22.184.127	443	15	2 kB	141	8	1 kB	7	806 bytes
172.16.1.89	49201	185.22.184.127	443	15	2 kB	142	8	1 kB	7	806 bytes
172.16.1.89	49202	185.22.184.127	443	15	2 kB	176	8	1 kB	7	806 bytes
172.16.1.89	49203	185.22.184.127	443	15	2 kB	180	8	1 kB	7	806 bytes
172.16.1.89	49204	185.22.184.127	443	23	7 kB	201	11	2 kB	12	5 kB
172.16.1.89	49205	185.22.184.127	443	23	7 kB	202	11	2 kB	12	5 kB
172.16.1.89	49206	185.22.184.127	443	15	2 kB	203	8	1 kB	7	806 bytes
172.16.1.89	49207	185.22.184.127	443	15	2 kB	204	8	1 kB	7	806 bytes
172.16.1.89	49214	185.22.184.127	443	15	2 kB	228	8	1 kB	7	806 bytes
172.16.1.89	49215	185.22.184.127	443	15	2 kB	239	8	1 kB	7	806 bytes
172.16.1.89	49217	185.22.184.127	443	15	2 kB	243	8	1 kB	7	806 bytes
172.16.1.89	49218	185.22.184.127	443	23	7 kB	244	11	2 kB	12	5 kB
172.16.1.89	49219	185.22.184.127	443	15	2 kB	246	8	1 kB	7	806 bytes
172.16.1.89	49220	185.22.184.127	443	15	2 kB	248	8	1 kB	7	806 bytes
172.16.1.89	49221	185.22.184.127	443	15	2 kB	249	8	1 kB	7	806 bytes
172.16.1.89	49222	185.22.184.127	443	15	2 kB	250	8	1 kB	7	806 bytes
172.16.1.89	49223	185.22.184.127	443	15	2 kB	251	8	1 kB	7	806 bytes
172.16.1.89	49224	185.22.184.127	443	23	7 kB	252	11	2 kB	12	5 kB

Conversation Settings		Ethernet - 20	IPv4 - 50	IPv6	TCP - 406	UDP - 140						
	Name resolution	Address A	Port A	Address B	Port B	Packets	Bytes	Stream ID	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A
	Absolute start time	172.16.1.141	49177	174.127.99.158	2017	2	120 bytes	307	1	66 bytes	1	54 bytes
	Display raw data	172.16.1.141	49177	174.127.99.158	2017	2	116 bytes	308	1	62 bytes	1	54 bytes
	Limit to display filter	172.16.1.141	49178	174.127.99.158	2017	2	120 bytes	309	1	66 bytes	1	54 bytes
		172.16.1.141	49178	174.127.99.158	2017	2	120 bytes	310	1	66 bytes	1	54 bytes
		172.16.1.141	49178	174.127.99.158	2017	2	116 bytes	311	1	62 bytes	1	54 bytes
		172.16.1.141	49179	174.127.99.158	2017	2	120 bytes	312	1	66 bytes	1	54 bytes
		172.16.1.141	49179	174.127.99.158	2017	2	120 bytes	313	1	66 bytes	1	54 bytes
		172.16.1.141	49179	174.127.99.158	2017	2	116 bytes	314	1	62 bytes	1	54 bytes
		172.16.1.141	49180	174.127.99.158	2017	2	120 bytes	315	1	66 bytes	1	54 bytes
		172.16.1.141	49180	174.127.99.158	2017	2	120 bytes	316	1	66 bytes	1	54 bytes
		172.16.1.141	49180	174.127.99.158	2017	2	116 bytes	317	1	62 bytes	1	54 bytes
		172.16.1.141	49181	174.127.99.158	2017	2	120 bytes	318	1	66 bytes	1	54 bytes
		172.16.1.141	49181	174.127.99.158	2017	2	120 bytes	319	1	66 bytes	1	54 bytes
		172.16.1.141	49181	174.127.99.158	2017	2	116 bytes	320	1	62 bytes	1	54 bytes
		172.16.1.141	49182	174.127.99.158	2017	2	120 bytes	321	1	66 bytes	1	54 bytes
		172.16.1.141	49182	174.127.99.158	2017	2	120 bytes	322	1	66 bytes	1	54 bytes
		172.16.1.141	49182	174.127.99.158	2017	2	116 bytes	323	1	62 bytes	1	54 bytes
		172.16.1.141	49183	174.127.99.158	2017	2	120 bytes	324	1	66 bytes	1	54 bytes
		172.16.1.141	49183	174.127.99.158	2017	2	120 bytes	325	1	66 bytes	1	54 bytes

IOC Observed	Port	Protocol
C2 beaconing — consistent 2KB/5KB packet pattern	49168	TCP
LDAP Active Directory enumeration — naming contexts queried	389	LDAP
Bidirectional file transfer between internal and external IPs	Multiple	TCP
C2 communication disguised as HTTPS traffic	443	TCP
Constant ACK/PSH-ACK heartbeat — automated beaconing	433	TCP
Ingress tool transfer delivered via ACK packets	433	TCP

POST request — suspected data exfiltration	8080	HTTP
--	------	------

Fake CRL file delivered from unrated external IP	80	HTTP
Mozilla user agent used to masquerade malicious traffic	8080	HTTP
File transfers with no filename — deliberate concealment	80/8080	HTTP

MITRE ATT&CK MAPPING

Technique ID	Technique name	Tactic
T1071.001	Application Layer Protocol: Web Protocols	Command & Control
T1571	Non-Standard Port (49168, 8080)	Command & Control
T1573.001	Encrypted Channel over HTTPS	Command & Control
T1008	Fallback Channels — multiple C2 servers	Command & Control
T1095	Non-Application Layer Protocol	Command & Control
T1001	Data Obfuscation via ACK tunneling	Command & Control
T1105	Ingress Tool Transfer	Command & Control

T1029	Scheduled Transfer	Exfiltration
T1041	Exfiltration Over C2 Channel	Exfiltration
T1048	Exfiltration Over Alternative Protocol	Exfiltration
T1069	Permission Groups Discovery	Discovery
T1018	Remote System Discovery	Discovery
T1087	Account Discovery via LDAP	Discovery
T1036	Masquerading — Mozilla user agent spoofing	Defense Evasion
T1553.004	Subvert Trust Controls: Install Root Certificate	Defense Evasion

MALWARE ASSOCIATION:

During investigation, IP address 153.149.153.41 was found to be associated with Emotet Trojan (Trojan.Win32) malware based on VirusTotal relational analysis.

Files associated with this IP were flagged by 39 to 60 security vendors

Malware classification: Trojan.Win32 / Emotet

No malicious files were directly extracted from the PCAP

Extracted payload hash returned no VirusTotal results — consistent with a custom or targeted variant

Note: A clean or undetected VirusTotal result does not confirm a file is safe. Network behaviour observed throughout this investigation strongly indicates malicious activity regardless of hash detection results.

CONCLUSION:

This PCAP investigation revealed a sophisticated multi-stage attack involving:

Multiple confirmed malicious C2 servers communicating with internal machines

Active Directory enumeration indicating post-exploitation reconnaissance

Data exfiltration via HTTP POST requests on non-standard ports

Defense evasion techniques including HTTPS masquerading, fake CRL delivery and Mozilla user agent spoofing

Emotet trojan infrastructure linked to the primary malicious IP

The attack pattern suggests the threat actor had already established a foothold inside the network before this PCAP capture began. The method of initial intrusion could not be determined from this PCAP alone and would require additional log sources such as firewall logs, email gateway logs or endpoint detection data.